University of Glasgow

Course Work

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UESTC ID: [redacted]

Name:Chen YiChi

University of Glasgow

PN Junction Diode - Lab 1

Coursework Aim:

This lab experiment aims to evaluate the characteristics of a PN junction diode in forward and reverse bias.

This lab is marked out of 100.

Course Learning Outcomes:

Upon successful completion of this assessment activity, students are expected to be able to:

- Explain and characterize the principle of operation of important electronic devices (pn-junction diodes).
- Sketch the current-voltage (I-V) characteristics of key electronic devices using appropriate equations and physical approximations.

Apparatus Required:

1. Bread Board.
2. Voltmeter.
3. Ammeter.
4. DC power supply.
5. PN junction Diode.
6. Computer with SPICE software (ORCAD/LTSPICE).

Background:

A PN junction diode is a two-terminal device consisting of a P-N junction formed using a semiconducting material such as Silicon. The P-type material has a high concentration of holes, and the N-type material has a high concentration of free electrons. Due to the concentration gradients on either side of the junction, excess holes will diffuse from the P to the N side, and electrons will diffuse from the N to the P. This process is known as diffusion. At equilibrium, the depletion region prevents further diffusion of carriers. Therefore, the output current is zero.

For the following exercises, choose a diode from the lab and make sure to note its model number. Based on this model number, download its datasheet. Aim to find the same model number from Spice's library (or a similar diode) and complete the following exercises:

Exercises:

Setup the PN diode in:

1) Forward bias: The P-type material should be connected to the positive terminal, while the N-type is connected to the negative terminal of a battery. No current flows across the diode when the applied voltage is below the barrier (or threshold) voltage. Make sure to note the forward-biased current and voltages in the table below. The forward current rises exponentially as the forward applied voltage increases beyond the threshold voltage. Compare your simulation and experimental results. [10 marks]

v(F) **simulation**	I(mA)
0.1	0
0.2	0
0.5	0.15
0.6	1.4
0.7	11.2
0.8	57

v(F)practical	I
0.14	0
0.3	0
0.52	0.1
0.6	0.6
0.63	1
0.71	3
1	24.2
1.2	40.5
1.33	52
1.4	59.5

1.6	**84.3**
1.7	**95.2**
1.81	**106**

Comparation: The difference between the simulation and the experimental results is very small.

2) Reverse bias: The P-type material is connected to the negative terminal, while N-type is connected to the positive terminal of a battery. Here, the size of the depletion region is increased and less current flows in the circuit. Compare your simulation and experimental results.
[10 marks]

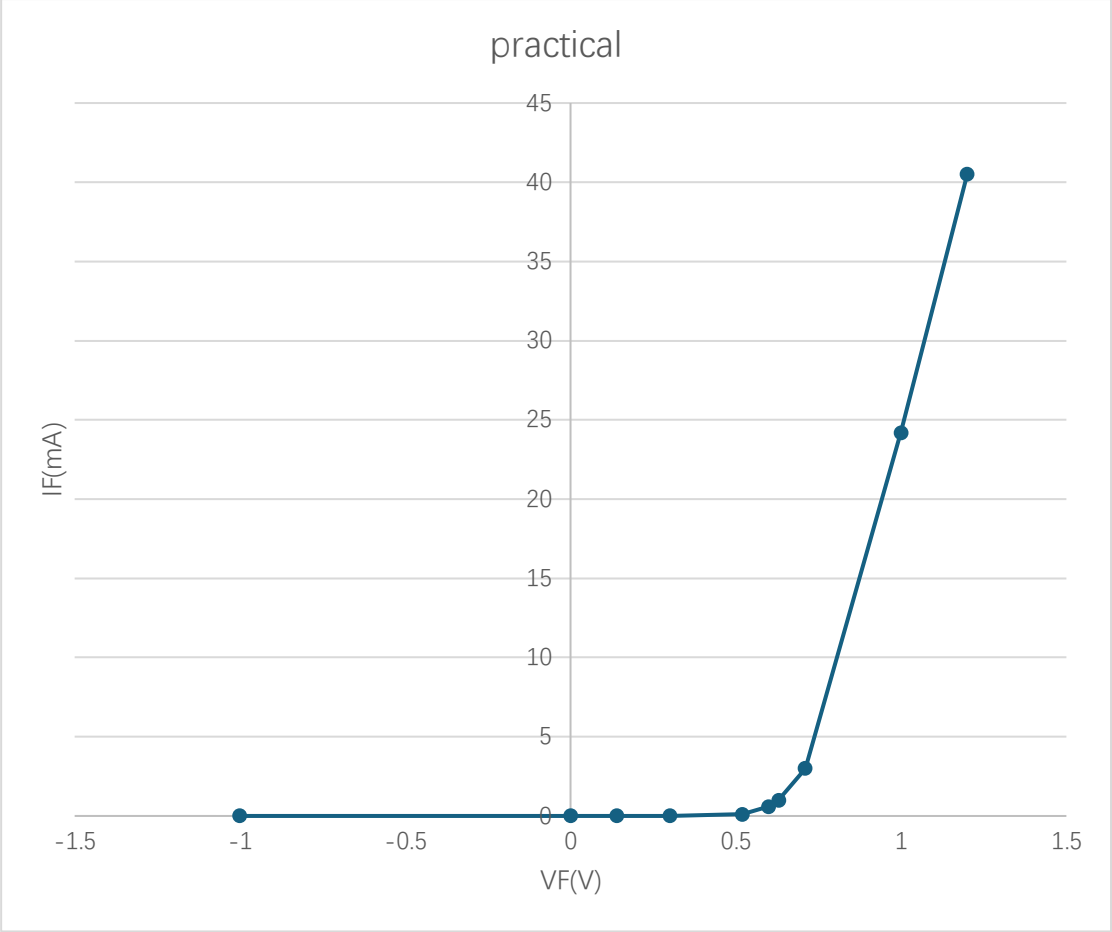
v(F)(simulation)	I(nA)
0	0
-0.1	-2.2
-0.2	-2.4
-0.3	-2.5
-0.4	-2.5

v(F)(practical)	I
0	0

Comparation: As for simulation results, the value had been around 2.5nA.

As for experimental results, the value could not be detected.

3) Sketch the IV-curve of the PN-junction diode, making sure to label all the major parameters, show your explanations on the IV-curve, and argue about Forward Bias and Reverse Bias characteristics and what conclusions you extract from the IV-curve? [40 mark]



1. Forward Bias Characteristics (Positive Voltage):

From the figure, the current began to increase significantly as the forward voltage exceeded a threshold voltage (~0.6V). Here's how the data aligned with the typical characteristics of a forward-biased diode:

- Threshold Voltage (~0.6V - 0.7V): The current started to increase at around 0.6V. This was close to the typical threshold voltage for a silicon diode, which was typically around 0.7V. Below this voltage, the current remained close to zero.
- Exponential Increase: After the threshold voltage (~0.6V), the current rose rapidly with an increase in voltage. For example:
    - At 0.63V, the current was 1 mA.
    - At 0.71V, the current jumped to 3 mA.
    - At 1.0V, the current was 24.2 mA.
    - At 1.33V, the current was 52 mA.
    - At 1.6V, the current reached 84.3 mA.
    - At 1.81V, the current reached 106 mA.

This demonstrated the exponential nature of the current in the forward bias region once the threshold voltage was surpassed.

2. Reverse Bias Characteristics (Negative Voltage):

- Reverse Saturation Current: The current remained close to zero at negative voltages, which indicated that there was little to no current flowing through the diode in the reverse

bias region (this matched the expected reverse leakage current, which was typically very small).

- **3.conclusion**

  Threshold Voltage: The threshold voltage in my experiment was around 0.6V to 0.7V, where the current started increasing.

  Exponential Growth: After the threshold voltage, the current grew exponentially with increasing voltage, showing that the diode was in the forward active region.

4) Assume that a PN junction diode consists of Silicon ($n_i = 1.5 \times 10^{10} cm^{-3}$) on both sides. The P-type is doped with Boron, and the N-type is doped with Phosphorus. Both sides are doped at the same concentration level of $1 \times 10^{15} cm^{-3}$. Employ appropriate formulas to solve:

   (i) the built-in voltage, (V_{bi}). [5 marks]

   (ii) the size of the depletion region, assuming $\epsilon_r = 11.7$. [10 marks]

1) $V_{bi} = \frac{KT}{q} \ln \frac{NDNA}{{n_i}^2} = \frac{1.38\times10^{-23}\times300K}{1.6} \ln \frac{(1\times10^{15} cm^{-3})^2}{1.5\times10^{10} cm^{-3\,2}} = 0.596\ V$

2) $W = \sqrt{\frac{2\epsilon 0 \epsilon r V bi}{q}\left(\frac{1}{NA} + \frac{1}{NB}\right)} = 0.000122m$

5) Consider a silicon PN junction with $\epsilon_r = 12$, $n_i = 1.5 \times 10^{10} cm^{-3}$, N_A = 1×10^{18} cm^{-3} and N_D = 1×10^{17} cm^{-3}, maintained at T = 300K. For zero bias, employ appropriate formulas to solve:

   (i) The built-in potential (V_{bi}). [5 marks]

   (ii) the depletion layer width (W). [10 marks]

   (iii) The maximum electric field (E(0)). [10 marks]

1) $V_{bi} = \frac{KT}{q} \ln \frac{NDNA}{{n_i}^2} = \frac{1.38\times10^{-23}\times300K}{1.6} \ln \frac{1\times1018\ cm{-}3\times1\times1017\ cm{-}3,}{1.5\times10^{10} cm^{-3\,2}} = 0.8727V$

2) $W = \sqrt{\frac{2\epsilon 0 \epsilon r V bi}{q}\left(\frac{1}{NA} + \frac{1}{NB}\right)} = 1.129\times10^{-5}m$

3) $E(0) = \frac{Vbi}{W} = 77298.4942 V/m$

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Bipolar Junction Transistor - Lab 2

Coursework Aim:

This lab activity aims to demonstrate the characteristics of an npn BJT as a function of both V_{BE} and V_{CE} .

This lab is marked out of 100.

Course Learning Outcomes:

Upon successful completion of this assessment activity, students are expected to be able to:

- Explain and characterize the principle of operation of important electronic devices (BJTs).
- Evaluate the changes in key electronic device properties as the operating conditions are changed.

Apparatus Required:

1. Bread Board.
2. Voltmeter.
3. Ammeter.
4. DC power supply.
5. BJT.
6. Computer with SPICE software (ORCAD/LTSPICE).

Background:

A bipolar junction transistor (BJT) can be thought of as a set of back-to-back diodes, where the operation of one diode can influence the currents in the other diode because the common region (the base, B) of the two diodes is very thin. The other two terminals are the collector, C, and the emitter, E. The forward DC gain, β_F, of a bipolar junction transistor is the ratio of the collector current-to-the base current when the transistor is biased in the forward active region.

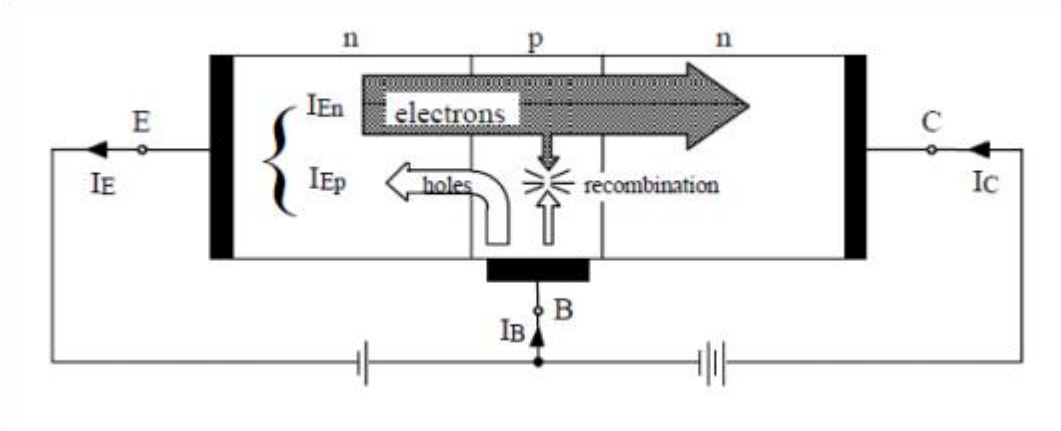


Figure 1: *npn* BJT in active mode of operation.

It is important to note that:

- $I_c \approx I_s \exp(\frac{V_{BE}}{V_T})$
- $I_c = \beta I_B$

Where:

- $I_s = Saturation\ Current$
- $V_T = Thermal\ Voltage = \frac{kT}{q} \approx 25mV$
- $\beta = Common\ Emitter\ current\ gain$

Thus, simulate the circuit according to Figure 2, with $R_B = 33k\Omega$ and $R_C = 1k\Omega$.

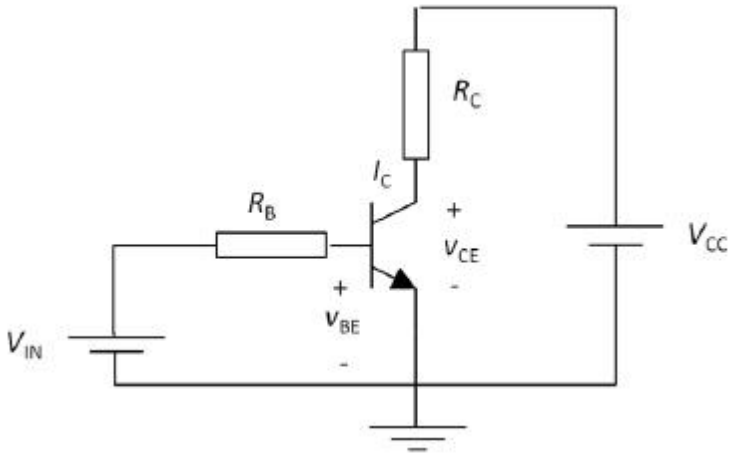


Figure 2: Connection of BJT in 'common-emitter' mode, since the emitter is connected to both input and output circuits.

Exercise 1 - I_c vs V_{BE}

[50 marks]

a) Download at least three datasheets for NPN general-purpose transistors. Evaluate and select one from them based on parameters of collector current, current gain, collector emitter saturation voltage, base emitter saturation voltage and thermal resistance. Can it be used for audio amplifiers? Justify your choice. [5]

The BC547B NPN transistor is **suitable** for audio amplifier applications based on the following key parameters from its datasheet:

Collector Current (IC): It can handle a continuous peak of 200mA, which is adequate for most audio applications.

Current Gain (hFE): With a minimum gain of 200 at 2mA collector current, it provides sufficient amplification for audio signals.

Collector Emitter Saturation Voltage (VCE(sat)): It has a low saturation voltage (<0.25V at 10mA, <0.60V at 100mA), which is beneficial for efficiency in audio amplifiers.

Base Emitter Saturation Voltage (VBE(sat)): The typical saturation voltage is 0.70V at 10mA and 0.90V at 100mA, which is acceptable for audio amplification.

Thermal Resistance (Rth(j-a)): With a value of 250°C/W, it can be used in audio amplifiers, but proper heat sinking may be required.

In summary, the BC547B's characteristics make it a viable choice for audio amplifiers, offering a good balance of current handling, gain, and efficiency.

b) Now use the **BC547B** transistor model, recall its datasheet and fill in the following blanks: [6]

- Collector current $I_c(T_A = 25°C)$: 100 mA
- Power dissipation P_D: 500 mW
- Collector-emitter breakdown voltage V_{CEO}: 45V

c) Measure the relation between the base-emitter voltage (V_{BE}) and the collector current (I_c) according to the table below. V_{CC} should be fixed to at least five different values in the range of 1V to 15V. [20]

V_{CC}	V_{IN}	I_c	V_{BE}	I_B	IC/IB

(V)	(V)	(mA)	(V)	(mA)	
15	0	15.09pA	497.17nV	-15.07pA	-1
	0.5	6.027µA	0.499	17.89nA	336.89
	1	3.315	0.667	10.08µA	328.87
	1.5	7.39	0.692	24.47µA	302
	2	10.93	0.706	39.20µA	278.83
	2.5	14	0.716	54.05µA	259.02
	3	14.83	0.718	69.12µA	214.55
	3.5	14.86	0.719	84.27µA	176.34
	4	14.87	0.719	99.41µA	149.58

V_{CC} (V)	V_{IN} (V)	I_c (mA)	V_{BE} (V)	I_B (mA)	IC/IB
10	0	10.09pA	332.17nV	-10.07pA	-1
	0.5	5.633µA	0.499	17.90nA	314.69
	1	3.097	0.667	10.08µA	307.24
	1.5	6.904	0.692	24.48µA	282.03
	2	9.804	0.705	39.25µA	249.78
	2.5	9.862	0.705	54.39µA	181.32
	3	9.879	0.706	69.53µA	142.08
	3.5	9.889	0.706	84.67µA	116.79
	4	9.896	0.706	99.81µA	99.15

V_{CC} (V)	V_{IN} (V)	I_c (mA)	V_{BE} (V)	I_B (mA)	IC/IB
5	0	5.091pA	167.17nV	-5.066pA	-1
	0.5	5.239µA	0.499	17.90nA	292.68
	1	2.88	0.667	10.08µA	285.71
	1.5	4.864	0.683	24.75µA	196.53

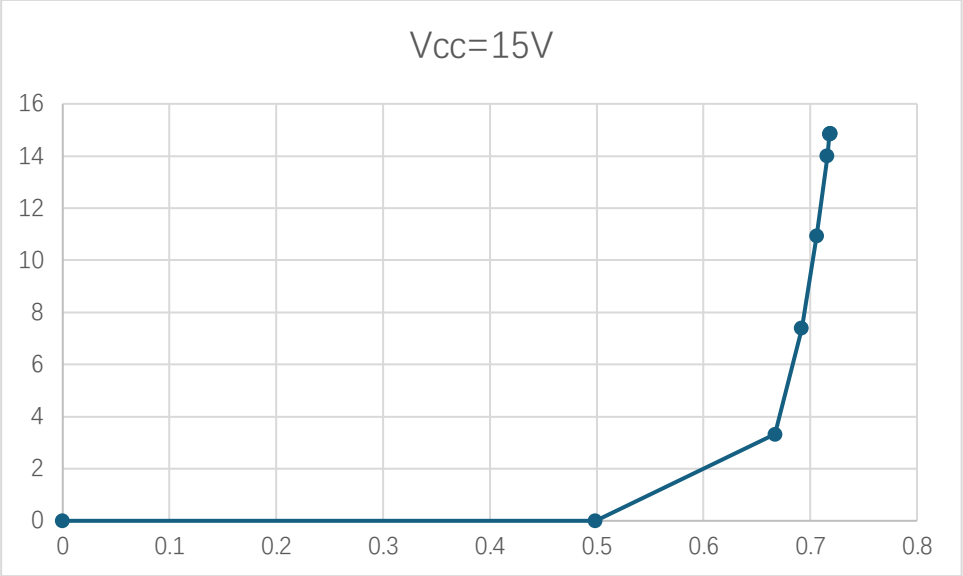
	2	4.894	0.684	39.88µA	122.72
	2.5	4.908	0.685	55.01µA	89.22
	3	4.916	0.685	70.14µA	70.09
	3.5	4.923	0.686	85.27µA	57.73
	4	4.928	0.687	0.1	49.23

V_{CC} (V)	V_{IN} (V)	I_c (mA)	V_{BE} (V)	I_B (mA)	IC/IB
	0	3.091pA	101.17nV	-3.066pA	-1
	0.5	5.081µA	0.499	17.90nA	283.85
	1	2.767	0.667	10.09µA	274.23
	1.5	2.9	0.669	25.18µA	115.17
3	2	2.918	0.67	40.29µA	72.42
	2.5	2.928	0.671	55.41µA	52.84
	3	2.935	0.672	70.54µA	41.61
	3.5	2.94	0.673	85.66µA	34.32
	4	2.944	0.674	0.101	29.15

V_{CC} (V)	V_{IN} (V)	I_c (mA)	V_{BE} (V)	I_B (mA)	IC/IB
	0	1.090pA	35.17nV	-1.066pA	-1
	0.5	4.924µA	0.499	17.90nA	275.08
	1	0.915	0.638	10.96µA	83.48
	1.5	0.941	0.642	26.01µA	36.18
1	2	0.952	0.644	41.08µA	23.17
	2.5	0.959	0.647	56.16µA	17.08
	3	0.963	0.649	71.24µA	13.52
	3.5	0.967	0.651	86.33µA	11.2
	4	0.969	0.653	0.101	9.59

d) Sketch the measured values in a I_c vs V_{BE} curve.

[10]



e) Examine the effect of V_{BE}, and use theoretical knowledge to support your analysis.

[9] **Effect of VBE on IC (Theoretical Explanation):**

- **Cutoff Region (VBE < 0.7V for Silicon BJTs):**
  When VBE was below the threshold (~0.7V for silicon), the base-emitter junction was not forward biased enough, and IC ≈ 0. The transistor was "off" and did not conduct (cutoff region).
- **Active Region (VBE > 0.7V):**
  When VBE exceeded the threshold, the base-emitter junction became forward biased. This allowed charge carriers (electrons in an NPN transistor) to flow from the emitter to the base, causing IC to increase.
- **Theoretical Relationship:** The relationship between VBE and IC was exponential in this region. The Shockley equation described this:

  $I_c = I_{b(}e^{\frac{VBE}{VT}} -1)$

  where:

  I_S was the saturation current (a very small constant),

  V_T was the thermal voltage (approximately 26mV at room temperature).

  This equation showed that I_C increased exponentially as V_{BE} increases.
- **Saturation Region (V_{BE} large, V_{CE} small):**
  As V_{BE} increased beyond a certain point, I_C eventually levelled off and became saturated. In this region, IC was no longer exponentially dependent on V_{BE}. Instead, I_C was primarily limited by external factors (such as the supply voltage or load resistance).

- **Key Theoretical Insights:**
- **Exponential Growth:** In the active region, the base-emitter voltage (V_{BE}) controlled the collector current (I_C) in an exponential manner, governed by the Shockley equation.
- **Thermal Voltage (V_T):** The thermal voltage (~26mV) played a crucial role in the rate at which IC increased for a given V_{BE}. A small increase in V_{BE} can cause a large increase in I_C.
- **Saturation:** When V_{BE} became very large, I_C reached a maximum value and saturates, entering the saturation region where the transistor behaved more like a switch than an amplifier.
- **Conclusion:**
  V_{BE} controls the transistor's state: Below 0.7V, the transistor was off (cutoff region). Between 0.7V and a larger value, I_C increased exponentially in the active region. After reaching a saturation point, I_C became largely independent of further increases in V_{BE} (saturation region).

Exercise 2 - I_c vs V_{CE}

[50 Marks]

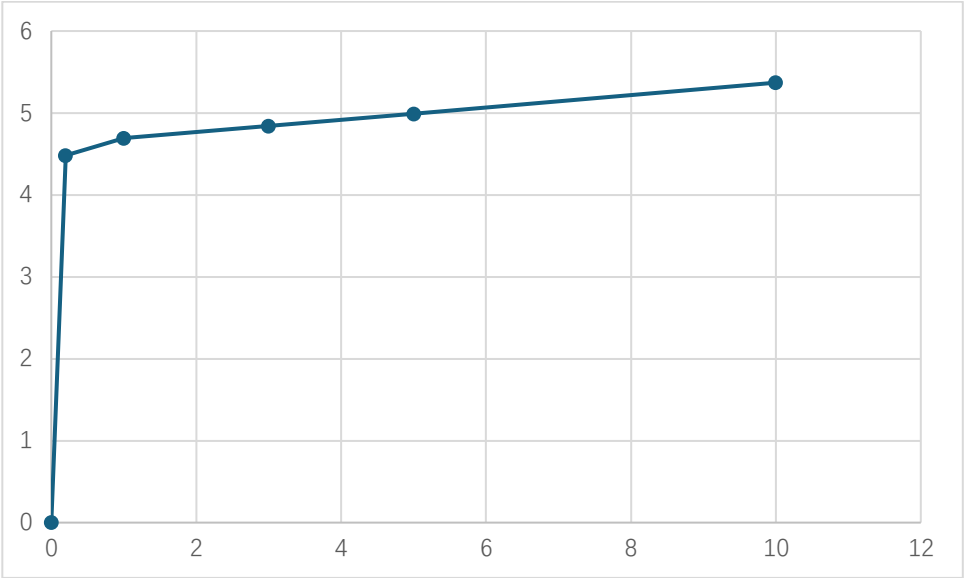
a) Analyze the effect of varying V_{CE} on I_c for the BC547B BJT with the input voltage V_{IN} kept constant. Fill out the table below.
[15]

- Set V_{CC} to 10V.
- Set V_{IN} so that V_{CE} = 5V (measure V_{BE} and compare to your previous measurement). Keep this value for V_{IN} for the entire measurement series.
- Vin=1.241v
- Adjust V_{CC} to set V_{CE} according to the table and measure I_c.
- Repeat the above steps for the bias points in the table.

V_{CE} (V)	I_c (mA)
0	0.00<0
0.2	4.48
1	4.69
3	4.84
5	4.99
10	5.37

b) Sketch the measured values in a I_c vs V_{CE} curve.

[10]



c) Recall the characteristics curve you learned earlier to discuss your results. Draw some conclusions and explain the reasons for this effect.

[10]

1. Initial Stage (V_{CE} = 0 to 0.2V):

When V_{CE} increased from 0V to 0.2V, IC rose from 0 to 4.48 mA, indicating that the transistor was transitioning from the cutoff region to the active region. This happened because as V_{CE} increased, the base-emitter junction became forward-biased, and the transistor started conducting.

2. Active Region (V_{CE} > 0.2V):

As V_{CE} increased further (from 0.2V to 10V), I_C increased gradually and then stabilized at about 5.37 mA at V_{CE} = 10V. In this region, the transistor was in the active region, where I_C was mainly controlled by the base current I_B and the base-emitter voltage V_{BE}. Changes in V_{CE} had a minor effect on I_C, because the transistor was already on and the current is mainly determined by the transistor's current gain.

3. Saturation Region (High V_{CE}):

Once V_{CE} reached a certain point, IC started to plateau, as seen from the data. Even as V_{CE} increased beyond this point, I_C no longer increased significantly. This indicated that the transistor had entered the saturation region, where further increases in V_{CE} did not affect I_C. The transistor was fully on, and I_C was now limited by external factors such as the load resistance.

Conclusions and Reasons:

In the active region, I_C increased with V_{CE}, but the rate of increase diminished as V_{CE} got higher. This was because the transistor was already conducting, and I_C was primarily controlled by V_{BE} and the base current.

In the saturation region, IC reached a maximum value and remained nearly constant, regardless of further increases in V_{CE}. This happened because the transistor was fully saturated, and I_C was primarily controlled by the external circuit (e.g., load resistor), not by V_{CE}.

The effect of V_{CE} on I_C was significant in the active region, but became negligible in the saturation region, which reflected the typical BJT output characteristics.

d) Use fundamental electronic laboratory skills to implement the circuit practically and repeat the above experiment. Record the actual measurement results and compare them with the simulation results. What conclusions can you draw from this?

V_{CE} (V)	I_c (mA)
0	0
0.2	4.43
1	4.54
3	4.73
5	4.80
10	5.17

Conclusion: The actual measured values of IC were generally lower than those simulated using LTspice.

[15]

JFET Characteristics - Lab 3

Coursework Aim:

This assessment activity aims to illustrate the characteristics of JFET as a function of both V_{DD} and V_{GS}.

This lab is marked out of 100.

Course Learning Outcomes:

Upon successful completion of this assessment activity, students are expected to be able to:

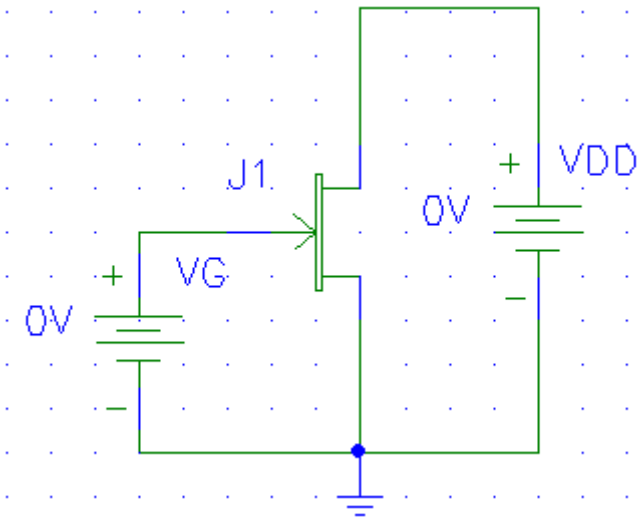
- Explain and characterize the principle of operation of important electronic devices (JFET).
- Sketch the current-voltage (I-V) characteristics of key electronic devices using appropriate equations and physical approximations.

Apparatus Required:

Computer with SPICE software (ORCAD/LTSPICE).

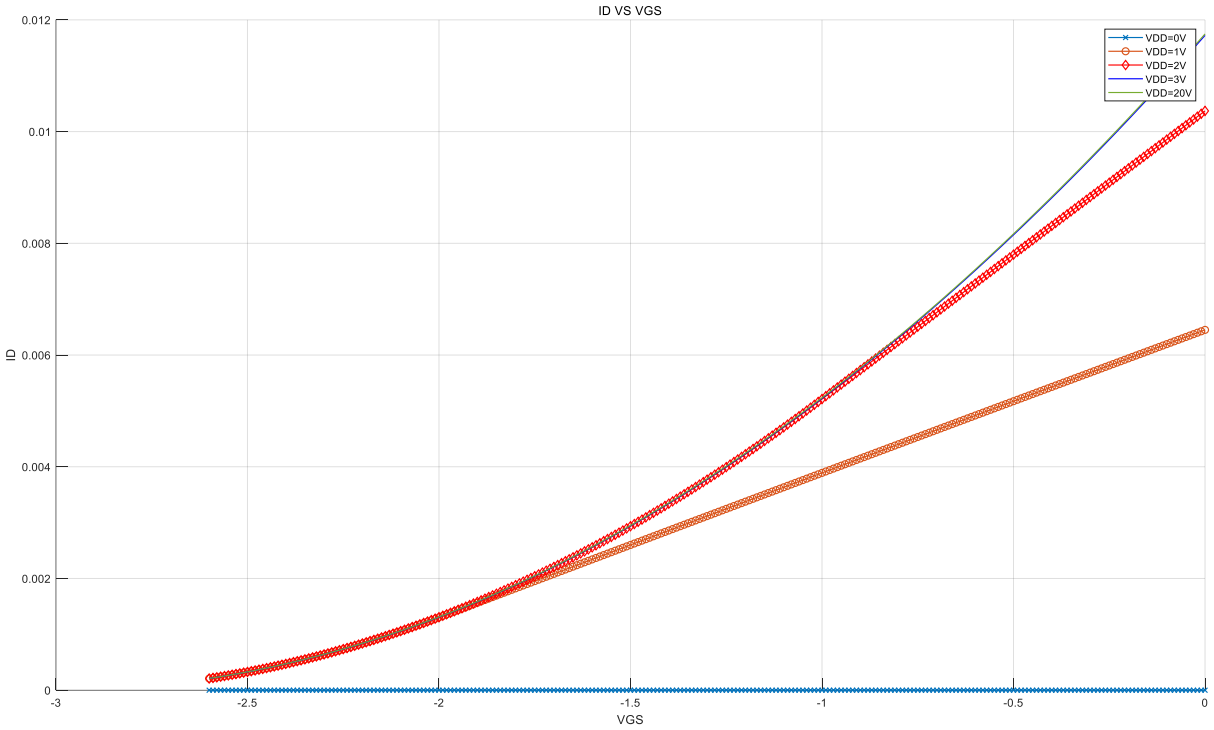
Background:

For this lab, we shall be using the 2N3819 JFET. Please go through the lab and indicate the missing parameters. Please upload this completed answers sheet via Moodle at the end of the lab. The circuit indicated in the schematic below will let us plot the I-V characteristics of the device.

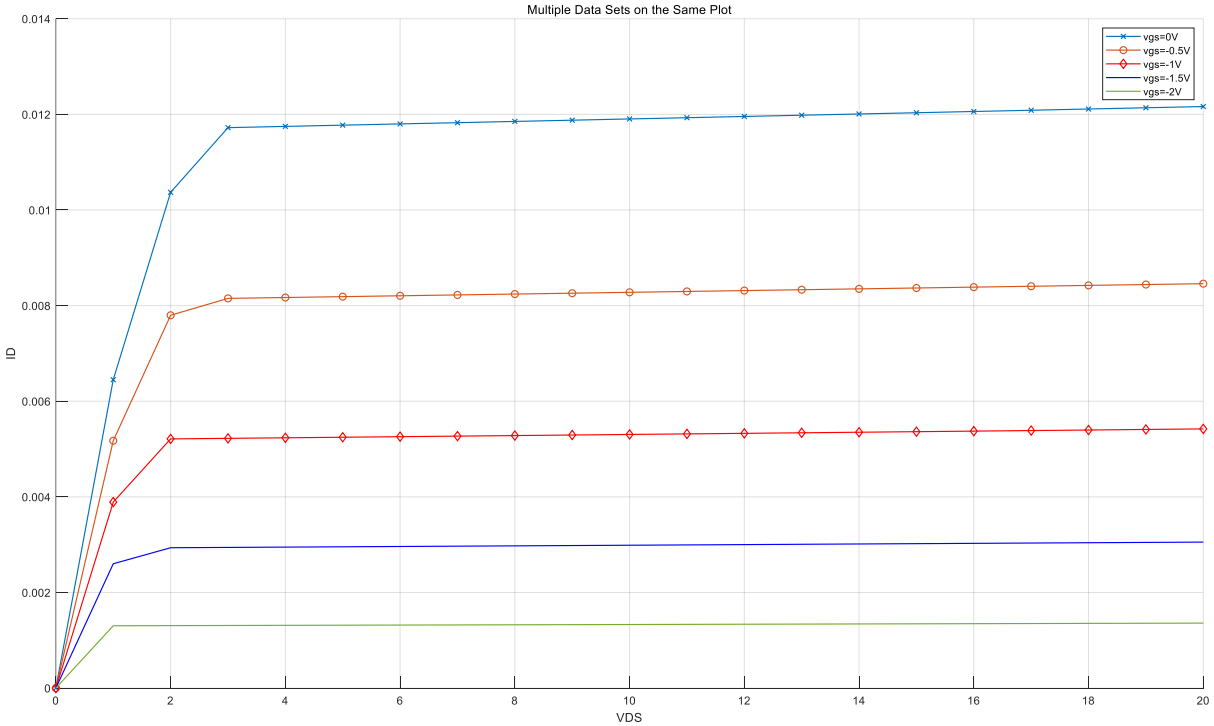


Exercises: You should construct a circuit that instructs **Spice** to plot the drain current as a function of V_{DD} (the drain voltage) and V_G (the gate voltage) for the prescribed limits below:

1) Illustrate the transfer characteristic I_D vs. V_{GS} for the 2N3819 n-channel JFET for V_{DD} = {0 to 20 V} and V_G = {-2.6 to 0 V} with clear labels. [10 marks]



2) Illustrate the transfer characteristic I_D vs. V_{DS} for the 2N3819 n-channel JFET for V_{GS} = {-2.5 to 0 V} with incremental steps of -0.5 V with clear labels. Please describe the relationship of I_D with V_{GS} for the JFET conclusively. [20 marks]



Relationship:

as VGS became more negative, the MOSFET became more conductive, and ID increased more quickly with increasing VDS.

The DC Operating Point and Load Line

Now that we have the characteristic I-V plot for the device, we can choose a DC operating point Q. We want the amplifier to work in a linear region to minimise the amount of signal distortion. The optimum region for the JFET is in the *constant current region.* We must avoid the region where $V_{GS} > 0$, and where V_{DS} exceeds the manufacturer's limit. Typically, we will choose an operating point Q that lies near the centre of the safe region. JFETs are typically set up to operate at *midpoint bias*. This means that the JFET is biased so that the drain-source voltage is approximately halfway between the supply voltage and earth potential.

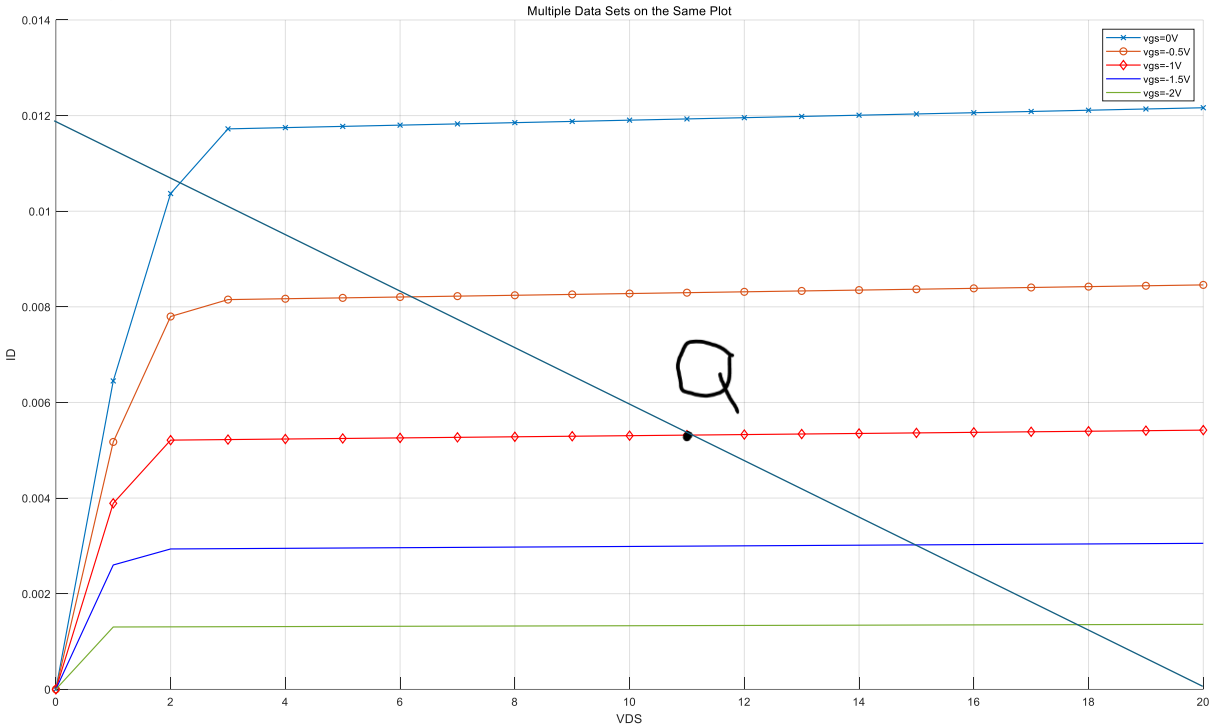
3) Describe how to choose the operating point, given that we will use a supply voltage of 20 V. Note the value of $I_D = 5.3mA$ and $V_{DS} = 11V$. [20 marks]

   The reason for choosing ID = 5.3 mA and VDS = 11V as the operating point for the JFET is to ensure the device operates in the current saturation region, which is crucial for minimizing signal distortion in amplifier applications.

Constructing the Load Line

Illustrate the IV curve at different V_{GS} values and identify the operating point as well as the DC load line.

4) The next step is to draw a line from the X intercept at the supply voltage point, through the operating point Q, all the way to the Y-axis. This is the DC load line, which governs the operation of the circuit and determines the value we choose for the drain resistor R_D. It has a slope - R_D. Please explain how to calculate the value of R_D? [15 marks]



RD=$\frac{Vsupply-VDS}{ID}$=1698Ω

Please explain what the voltage of the battery is required to bias the gate. [10 marks]

Voltage of battery=V_G=-1V

Response to a Small Signal (0.5 V) p-p sine wave

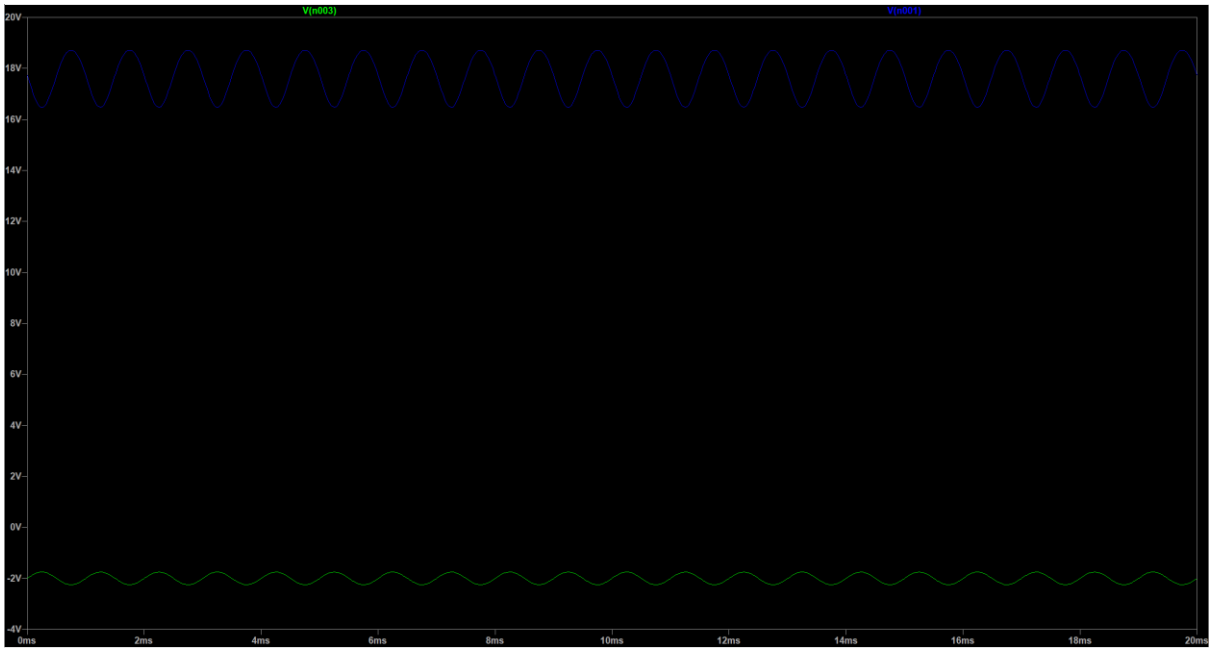
5) Apply a sine wave of 0.25 V peak.

   a) Illustrate the input and output waveforms and make sure to label all the major components.

   [10 marks]

   b) Please explain what is the value of voltage gain due to the JFET.

   [15 marks]



Voltage gain=$\frac{Vd}{Vg}$=-$\frac{VDmax-VDmin}{VGmac-VGmin}$≈-2

The negative voltage gain in your plot indicates that my **JFET amplifier** is an **inverting amplifier**. In this configuration, the output signal is **180° out of phase** with the input signal, meaning that as the input increases, the output decreases, and vice versa. This phase inversion is typical of a **common-source amplifier**, where the input is applied to the gate, and the output is taken from the drain. The negative gain reflects this phase shift, not a loss of amplification.